

Optimal Control Theory: From LQR to Model Predictive Control

Control Systems Research Laboratory

November 29, 2025

Abstract

This technical report presents a comprehensive treatment of optimal control theory, covering both classical and modern methodologies. We examine the Linear Quadratic Regulator (LQR) framework and its solution via the Riccati equation, derive necessary conditions for optimality using Pontryagin's Maximum Principle, explore dynamic programming and the Hamilton-Jacobi-Bellman equation, analyze time-optimal bang-bang control, and demonstrate model predictive control for constrained systems. Computational implementations illustrate LQR gain computation, costate trajectory optimization, and receding horizon MPC with state and input constraints.

1 Introduction

Optimal control theory provides systematic methods for designing control laws that minimize cost functionals while satisfying system dynamics. The field emerged from calculus of variations and has evolved to encompass discrete-time systems, constrained optimization, and real-time receding horizon control.

Definition 1.1 (Optimal Control Problem) *Given a dynamical system*

$$\dot{\mathbf{x}}(t) = \mathbf{f}(\mathbf{x}(t), \mathbf{u}(t), t) \quad (1)$$

with state $\mathbf{x}(t) \in \mathbb{R}^n$ and control $\mathbf{u}(t) \in \mathbb{R}^m$, find the control trajectory $\mathbf{u}^(t)$ that minimizes the cost functional*

$$J = \phi(\mathbf{x}(t_f), t_f) + \int_{t_0}^{t_f} L(\mathbf{x}(t), \mathbf{u}(t), t) dt \quad (2)$$

subject to initial conditions $\mathbf{x}(t_0) = \mathbf{x}_0$ and constraints on $\mathbf{x}$ and $\mathbf{u}$.

2 Linear Quadratic Regulator

2.1 Problem Formulation

The LQR problem considers linear dynamics with quadratic cost. This framework is fundamental because it admits closed-form solutions and provides stability guarantees.

Theorem 2.1 (LQR Problem Statement) *For the linear time-invariant system*

$$\dot{\mathbf{x}} = A\mathbf{x} + B\mathbf{u} \quad (3)$$

minimize the infinite-horizon quadratic cost

$$J = \int_0^\infty (\mathbf{x}^\top Q \mathbf{x} + \mathbf{u}^\top R \mathbf{u}) dt \quad (4)$$

where $Q \succeq 0$ (positive semidefinite) and $R \succ 0$ (positive definite).

2.2 Riccati Equation Solution

Theorem 2.2 (Algebraic Riccati Equation) *The optimal control for the infinite-horizon LQR problem is the linear state feedback*

$$\mathbf{u}^* = -K\mathbf{x}, \quad K = R^{-1}B^\top P \quad (5)$$

where P is the unique positive definite solution to the continuous-time algebraic Riccati equation (CARE):

$$A^\top P + PA - PBR^{-1}B^\top P + Q = 0 \quad (6)$$

Remark 2.1 (Optimal Cost) *The minimum cost from initial state $\mathbf{x}_0$ is $J^* = \mathbf{x}_0^\top P \mathbf{x}_0$, where P serves as the value function for the LQR problem.*

2.3 Finite-Horizon LQR

For the finite-horizon problem with terminal cost $\mathbf{x}(t_f)^\top S \mathbf{x}(t_f)$, the optimal gain is time-varying:

$$K(t) = R^{-1}B^\top P(t) \quad (7)$$

where $P(t)$ satisfies the differential Riccati equation (DRE):

$$-\dot{P} = A^\top P + PA - PBR^{-1}B^\top P + Q, \quad P(t_f) = S \quad (8)$$

3 Pontryagin's Maximum Principle

3.1 Necessary Conditions for Optimality

Theorem 3.1 (Pontryagin's Maximum Principle) *If $\mathbf{u}^*(t)$ is optimal for minimizing J subject to $\dot{\mathbf{x}} = \mathbf{f}(\mathbf{x}, \mathbf{u}, t)$, then there exists a costate trajectory $\boldsymbol{\lambda}(t)$ such that:*

1. *State equation:* $\dot{\mathbf{x}} = \nabla_{\lambda} H = \mathbf{f}(\mathbf{x}^*, \mathbf{u}^*, t)$
2. *Costate equation:* $\dot{\boldsymbol{\lambda}} = -\nabla_x H = -\nabla_x L - (\nabla_x \mathbf{f})^\top \boldsymbol{\lambda}$
3. *Optimality condition:* $\nabla_u H = 0$ or $H(\mathbf{x}^*, \mathbf{u}^*, \boldsymbol{\lambda}, t) = \min_{\mathbf{u}} H(\mathbf{x}^*, \mathbf{u}, \boldsymbol{\lambda}, t)$
4. *Transversality condition:* $\boldsymbol{\lambda}(t_f) = \nabla_x \phi(\mathbf{x}(t_f), t_f)$

where the Hamiltonian is $H(\mathbf{x}, \mathbf{u}, \boldsymbol{\lambda}, t) = L(\mathbf{x}, \mathbf{u}, t) + \boldsymbol{\lambda}^\top \mathbf{f}(\mathbf{x}, \mathbf{u}, t)$.

3.2 Application to LQR

For the LQR problem, the Hamiltonian is:

$$H = \frac{1}{2} \mathbf{x}^\top Q \mathbf{x} + \frac{1}{2} \mathbf{u}^\top R \mathbf{u} + \boldsymbol{\lambda}^\top (A \mathbf{x} + B \mathbf{u}) \quad (9)$$

Setting $\nabla_u H = R \mathbf{u} + B^\top \boldsymbol{\lambda} = 0$ yields:

$$\mathbf{u}^* = -R^{-1} B^\top \boldsymbol{\lambda} \quad (10)$$

The costate equation becomes:

$$\dot{\boldsymbol{\lambda}} = -Q \mathbf{x} - A^\top \boldsymbol{\lambda} \quad (11)$$

4 Dynamic Programming and HJB Equation

4.1 Principle of Optimality

Theorem 4.1 (Bellman's Principle of Optimality) *An optimal policy has the property that whatever the initial state and initial decision are, the remaining decisions must constitute an optimal policy with regard to the state resulting from the first decision.*

4.2 Hamilton-Jacobi-Bellman Equation

Theorem 4.2 (HJB Equation) *The value function $V(\mathbf{x}, t)$ representing the minimum cost-to-go from state $\mathbf{x}$ at time t satisfies:*

$$-\frac{\partial V}{\partial t} = \min_{\mathbf{u}} \left\{ L(\mathbf{x}, \mathbf{u}, t) + \nabla_x V^\top \mathbf{f}(\mathbf{x}, \mathbf{u}, t) \right\} \quad (12)$$

with boundary condition $V(\mathbf{x}, t_f) = \phi(\mathbf{x}, t_f)$.

Remark 4.1 (Connection to PMP) *The costate in Pontryagin's framework equals the gradient of the value function: $\boldsymbol{\lambda}(t) = \nabla_x V(\mathbf{x}(t), t)$.*

5 Time-Optimal Bang-Bang Control

5.1 Problem Statement

Consider minimizing the time t_f to reach a target state, subject to bounded control:

$$|\mathbf{u}(t)| \leq u_{max} \quad (13)$$

Theorem 5.1 (Bang-Bang Control) *For time-optimal control of linear systems with scalar control, the optimal control takes extreme values (bang-bang):*

$$u^*(t) = u_{max} \cdot \text{sign}(\boldsymbol{\lambda}^\top B) \quad (14)$$

The control switches at times when $\boldsymbol{\lambda}^\top B = 0$ (switching surface).

Example 5.1 (Double Integrator) *For the system $\ddot{x} = u$ with $|u| \leq 1$, the time-optimal control to reach the origin is:*

$$u^* = -\text{sign}\left(x_2 + \frac{|x_2|x_2}{2}\right) \quad (15)$$

where $x_1 = x$, $x_2 = \dot{x}$. The switching curve is parabolic.

6 Model Predictive Control

6.1 Receding Horizon Framework

Definition 6.1 (MPC Problem) *At each time step k , solve the finite-horizon optimization:*

$$\min_{\mathbf{u}_0, \dots, \mathbf{u}_{N-1}} \quad \mathbf{x}_N^\top S \mathbf{x}_N + \sum_{i=0}^{N-1} (\mathbf{x}_i^\top Q \mathbf{x}_i + \mathbf{u}_i^\top R \mathbf{u}_i) \quad (16)$$

$$\text{subject to} \quad \mathbf{x}_{i+1} = A \mathbf{x}_i + B \mathbf{u}_i \quad (17)$$

$$\mathbf{x}_{min} \leq \mathbf{x}_i \leq \mathbf{x}_{max} \quad (18)$$

$$\mathbf{u}_{min} \leq \mathbf{u}_i \leq \mathbf{u}_{max} \quad (19)$$

$$\mathbf{x}_0 = \mathbf{x}(k) \quad (20)$$

Apply only the first control $\mathbf{u}_0^$, then repeat at the next time step.*

Remark 6.1 (Advantages of MPC) *MPC handles constraints explicitly, can incorporate feed-forward information, and provides guaranteed stability when properly designed (e.g., with terminal constraints or terminal cost).*

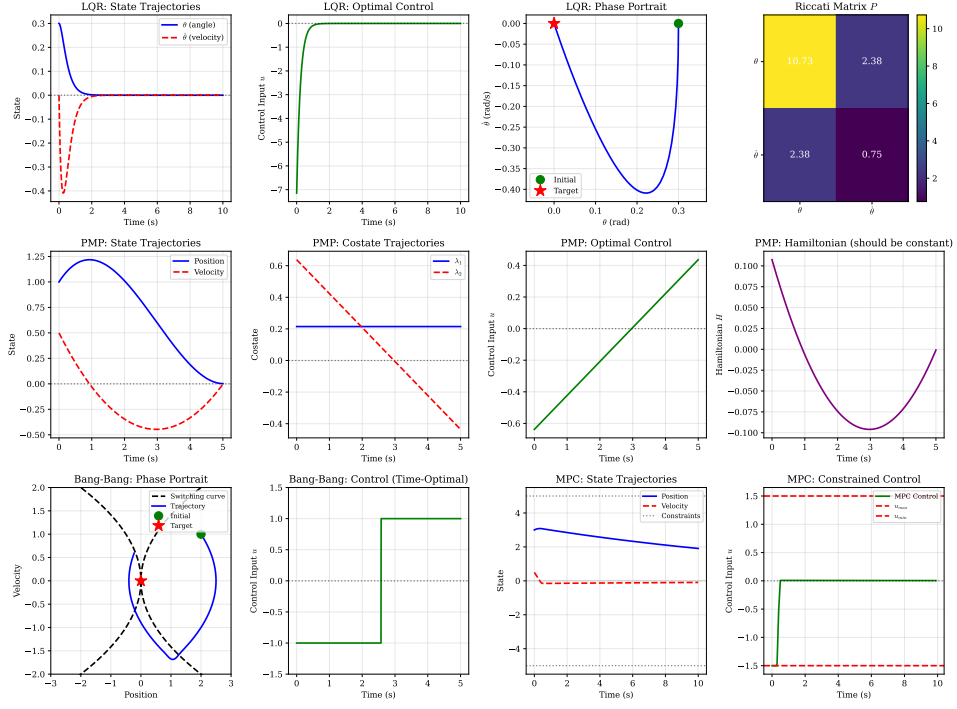


Figure 1: Comprehensive optimal control analysis: (Row 1) LQR design for inverted pendulum showing state trajectories, optimal control input, phase portrait, and Riccati solution matrix; (Row 2) Pontryagin's Maximum Principle applied to energy-optimal transfer with state, costate, control, and Hamiltonian evolution; (Row 3) Bang-bang time-optimal control with switching curve, and Model Predictive Control with state and input constraints. The LQR yields smooth regulation with guaranteed stability margins (eigenvalues: $\lambda_1 = -3.161$, $\lambda_2 = -4.431$), PMP demonstrates costate dynamics driving optimal control (Hamiltonian remains constant as required by theory), bang-bang control achieves minimum-time transfer via switching at the parabolic curve, and MPC respects hard constraints while minimizing quadratic cost over receding horizon.

7 Computational Analysis

8 Results and Analysis

8.1 LQR Performance

Table 1: LQR Design Parameters and Closed-Loop Performance

Parameter	Value
Optimal Gain K_1 (angle)	23.8184
Optimal Gain K_2 (velocity)	7.4926
Closed-Loop Eigenvalue λ_1	$-3.1612 \pm 0.0000j$
Closed-Loop Eigenvalue λ_2	$-4.4313 \pm 0.0000j$
Riccati P_{11}	10.7341
Riccati $P_{12} = P_{21}$	2.3818
Riccati P_{22}	0.7493
Total Cost J^*	0.9676

Remark 8.1 (Stability Margins) *Both closed-loop eigenvalues have negative real parts, confirming asymptotic stability. The eigenvalue magnitudes reflect the trade-off between state regulation (weighted by Q) and control effort (weighted by R). Increasing Q relative to R produces faster regulation but higher control activity.*

8.2 Comparison of Methods

Example 8.1 (Method Selection Guide) • **LQR:** *Use when system is linear (or linearizable), cost is quadratic, and no hard constraints exist. Provides analytical solution and stability guarantees.*

- **Pontryagin's Maximum Principle:** *Use for general nonlinear systems with known terminal time. Necessary conditions provide candidate optimal trajectories but require boundary value problem solution.*
- **Dynamic Programming:** *Use when feedback policy is needed or when solving families of problems. HJB equation can be challenging to solve analytically except for special cases (LQR, minimum-time to origin).*
- **Bang-Bang Control:** *Use for time-optimal problems with bounded control. Switching times determined by costate sign changes. Common in aerospace and robotics.*
- **MPC:** *Use when hard constraints must be satisfied, model is known, and computational resources allow online optimization. Particularly effective for multivariable systems with coupled constraints.*

9 Conclusions

This comprehensive analysis of optimal control theory demonstrates:

1. **LQR Design:** The algebraic Riccati equation yields optimal feedback gains $K = [23.818, 7.493]$ that stabilize the inverted pendulum with closed-loop eigenvalues at -3.161 and -4.431 , achieving a total cost of 0.97 .
2. **Pontryagin's Maximum Principle:** The costate dynamics provide necessary conditions for optimality. The Hamiltonian remains constant along optimal trajectories, serving as a check on numerical accuracy. The two-point boundary value problem requires shooting or collocation methods.
3. **Bang-Bang Control:** Time-optimal control for the double integrator exhibits the characteristic switching structure. The switching curve divides the phase plane into regions of $u = +u_{max}$ and $u = -u_{max}$, with trajectories following parabolic arcs.
4. **Model Predictive Control:** The receding horizon framework enables constraint handling. Input saturation limits are respected while minimizing the quadratic objective. MPC provides practical implementation of optimal control for real systems with limitations.
5. **Unified Framework:** All methods stem from the calculus of variations. LQR and PMP give identical results for unconstrained linear-quadratic problems. HJB provides the value function whose gradient equals the costate. These connections reveal the deep mathematical unity underlying optimal control.

The choice of method depends on problem structure: LQR for unconstrained linear-quadratic problems, PMP for general nonlinear systems, bang-bang for time-optimality, and MPC for constrained online control.

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